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# EXPERIMENT – 29

## Simulation and Analysis of GSM Communication System with Time Division Multiple Access (TDMA) Using MATLAB

**PROGRAM:**

```
clc;
clear;
close all;
% Four GSM Users
u1 = [1 0 1 1];
u2 = [0 1 0 1];
u3 = [1 1 0 0];
u4 = [0 0 1 1];
% TDMA Frame
tdma = [u1 u2 u3 u4];
% Display
disp('TDMA Frame:');
disp(tdma);
% Plot
figure;
subplot(2,1,1)
stem(tdma,'filled')
title('GSM TDMA Frame')
xlabel('Time Slot')
ylabel('Bit')
grid on
subplot(2,1,2)
stairs(tdma,'LineWidth',1.5)
```

```
title('TDMA Signal')
xlabel('Time Slot')
ylabel('Amplitude')
grid on;
```

**OUTPUT:**

